

AI-DRIVEN FOREST MONITORING SYSTEM FOR DEFORESTATION DETECTION AND EARLY FOREST-FIRE RISK PREDICTION USING DEEP LEARNING AND GEOSPATIAL HEATMAPS

MICRO PROJECT SYNOPSIS

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by

Name	Enrollment No
Krish Gupta	20011502724
Dev Khanna	35911502724
Piyush Gautam	36511502724

Guided by

Dr. Gargi Mishra
Assistant Professor



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
BHARATI VIDYAPEETH'S COLLEGE OF ENGINEERING
(AFFILIATED TO GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY, DELHI)
DELHI – 110063

1. Introduction

Forests regulate the global climate through carbon sequestration, sustain biodiversity, and support the livelihoods of hundreds of millions of people. Their continued loss is therefore not merely an ecological concern but a direct threat to climate stability, food and water security, and long-term socio-economic resilience.

Forests worldwide face sustained pressure from agricultural expansion, urbanisation, illegal logging, and infrastructure development, while forest fires are growing more frequent and intense due to rising temperatures and dry-biomass accumulation. Deforestation and fire risk are not independent: forest-cover loss alters micro-climatic conditions and fuel-load distribution, which in turn affects fire susceptibility — so monitoring the two together gives a more complete picture of forest vulnerability than monitoring either in isolation.

Conventional forest monitoring, based on ground surveys and manual interpretation of aerial or satellite photographs, is slow, labour-intensive, and difficult to scale across large or inaccessible terrain, often reporting damage only after it has occurred. The growing availability of free, high-revisit satellite missions such as Sentinel-2 and Landsat, combined with advances in deep learning for image segmentation (e.g., U-Net) and ensemble machine learning for tabular prediction (e.g., Random Forest, XGBoost), now enables automated, data-driven monitoring at a scale and consistency manual interpretation cannot match.

Predicting fire probability before ignition offers a fundamentally different, complementary capability to existing hotspot-alert services: it allows forest departments to pre-position resources, schedule controlled burns, and issue advisories ahead of elevated-risk periods. Combined with continuous deforestation monitoring, such a system can highlight regions that are simultaneously losing forest cover and becoming more fire-prone — valuable for prioritising limited conservation and enforcement resources.

This project proposes an integrated AI-based forest-monitoring framework combining (i) deep-learning deforestation and forest-cover change detection from multi-temporal satellite imagery, and (ii) machine-learning fire-risk prediction from historical fire records, vegetation indices, and weather variables. Both outputs are fused into a geospatial forest-vulnerability

visualisation intended as a decision-support and early-warning prototype, aligned with UN Sustainable Development Goal 13 (Climate Action) and Goal 15 (Life on Land).

2. Problem Statement

Existing approaches to forest monitoring and fire-risk assessment suffer from structural limitations that reduce their usefulness for timely, scalable decision-making:

- Manual interpretation of aerial or satellite photographs cannot be repeated often enough to capture rapid changes in forest cover, while ground-based patrolling frequently reports deforestation events weeks or months after they occur.
- Continuous monitoring across large, topographically difficult terrain is expensive to scale in terms of personnel, transport, and equipment.
- Deforestation monitoring and fire-risk assessment are typically run as separate, fragmented workflows with different tools and institutional owners, preventing analysts from observing the relationship between forest loss and fire susceptibility in one view.
- Meteorological, vegetation, and historical fire-occurrence data are often available in disparate formats from different agencies, requiring substantial manual effort to harmonise before predictive modelling is possible.

The central technical problem this project addresses is: how can artificial intelligence and deep learning be used to detect forest-cover changes from satellite imagery and combine this with environmental and historical fire data to estimate fire risk and generate interpretable geospatial heatmaps for early-warning and forest-management decision support? This requires an integrated pipeline that (a) localises forest-cover loss from multi-temporal imagery, (b) models fire-risk probability from heterogeneous tabular data, and (c) fuses both outputs into a single, spatially interpretable visualisation usable without specialised remote-sensing expertise.

3. Motivation

The motivation for this project stems from the convergence of three factors: the increasing accessibility of free, high-quality satellite data; the demonstrated maturity of deep learning for geospatial image analysis; and the pressing need for scalable, low-cost environmental monitoring tools amid accelerating climate change.

- Environmental monitoring at scale: satellite-based AI can repeatedly cover large, remote forest areas at a fraction of the cost of manual surveys.
- Transferability: a model trained on one forest region can, in principle, be fine-tuned to other geographies using publicly available Sentinel-2 or Landsat coverage.
- Early-warning value: fire-risk prediction ahead of the fire season gives forest managers lead time for preventive action that active-fire detection, by definition, cannot provide.
- Data-driven decision support: converting raw imagery and tabular records into an interpretable heatmap supports evidence-based prioritisation of patrolling and conservation resources.
- Accessible AI for public good: relying only on open datasets (Sentinel-2, Landsat, NASA FIRMS, public weather archives) keeps the system reproducible and extensible without licensing constraints.

4. Literature Review

This section reviews research relevant to the project's two core modules — deep-learning deforestation/change detection and machine-learning fire-risk prediction — along with work on geospatial decision-support visualisation, and concludes with a research-gap analysis.

4.1 Deep Learning for Deforestation and Forest-Cover Change Detection

The U-Net encoder–decoder architecture of Ronneberger et al. [1], originally developed for biomedical segmentation, has become the foundational backbone for much deforestation-mapping research because its skip connections produce pixel-accurate masks from limited training data. Bragagnolo et al. [3] applied U-Net to Landsat-8 imagery of the Amazon, achieving high spatial precision but reduced accuracy at mixed land-cover boundaries; John and Zhang [4] added attention-gated skip connections to better capture small-scale deforestation, at the cost of higher training complexity.

Alzu'bi and Alsmadi [2] demonstrated that a modified encoder–decoder approach generalises to Jordanian forests beyond the Amazon-dominated literature, while Isaienkov et al. [5] showed that frequent Sentinel-2 revisits reduce false positives from seasonal vegetation change. Maretto et al. [6] modelled the temporal dimension of Amazon deforestation explicitly, improving boundary delineation; Wang et al. [7] proposed SiamHRNet-OCR to preserve high-resolution

features for detecting small, irregular clearings; and Fodor and Conde [8] fused optical and radar sources (Sentinel-1/2, Landsat-8) to improve robustness to cloud cover in tropical regions.

4.2 Machine Learning and Deep Learning for Forest-Fire Risk Prediction

Jain et al. [9] categorised wildfire-management ML literature into distinct problem classes, explicitly separating pre-ignition fire-occurrence/susceptibility prediction — the class this project adopts — from active-fire detection. Michael et al. [10] showed that NDVI-based vegetation-dryness metrics substantially improve fire-risk mapping accuracy, with XGBoost outperforming Logistic Regression and Random Forest on Mediterranean woodlands. Singh and Jeganathan [11] applied ensemble ML to twenty years of MODIS fire data across central India, finding that NDVI and rainfall reduce fire probability while their negative anomalies amplify it.

Muhammad et al. [12] compared Logistic Regression, Random Forest, SVM, and XGBoost for fire-susceptibility mapping in Pakistan, again finding tree-based ensembles superior, with NDVI, land-surface temperature, and population density as top predictors; Shahzad et al. [13] reported similar ensemble-model superiority using the MODIS Global Fire Atlas. Ejaz and Choudhury [14] surveyed the end-to-end wildfire-risk ML pipeline and identified data sparsity, cross-region generalisability, and interpretability as open challenges that directly informed this synopsis's evaluation and limitations sections.

4.3 Geospatial Visualisation and Decision-Support Systems

GIS-based dashboards such as those used by the USDA Forest Service integrate weather, fuel, and fire-history layers into interactive risk-viewer tools, and research-oriented web-GIS tools combine fuel maps with socio-economic risk models. A recurring limitation across this literature is that operational dashboards often present a single deterministic risk category without conveying model confidence or the drivers behind a prediction — a gap this project addresses by reporting risk categories alongside the indicators driving each one.

4.4 Summary of Reviewed Literature

Study	Data / Source	Method	Key Finding	Limitation
Bragagnolo et al. [3] (2021)	Landsat-8, Amazon basin	U-Net semantic segmentation	High spatial precision for forest-cover change masks	Reduced accuracy on mixed-boundary / cloud-affected pixels
John & Zhang [4] (2022)	Satellite sensor imagery	Attention-based U-Net	Improved detection of small-scale deforestation	Higher training complexity and compute cost
Alzu'bi & Alsmadi [2] (2022)	Landsat-8, Jordan (2010–2020)	Modified encoder–decoder segmentation	New benchmark dataset; generalises beyond Amazon	Small, region-specific extent limits generalisation
Isaenkov et al. [5] (2021)	Sentinel-2 time series, Ukraine	Deep-learning change detection	Frequent revisit reduces false positives from seasonal change	Requires dense temporal image stacks
Maretto et al. [6] (2021)	Multitemporal imagery, Amazon	Spatio-temporal deep learning	Better deforestation-polygon boundary delineation	Higher data and compute requirements
Wang et al. [7] (2023)	High-resolution imagery	SiamHRNet-OCR	Improved detection of small/irregular clearings	High-resolution imagery not always freely available
Fodor & Conde [8] (2023)	Sentinel-1/2, Landsat-8, MODIS	Multimodal CNN	Optical + radar fusion improves cloud robustness	Complex multi-sensor pipeline
Michael et al. [10] (2021)	Satellite NDVI time series, Greece	LR / RF / XGBoost	NDVI dryness trend improves fire-risk mapping	Tuned to Mediterranean vegetation dynamics
Singh & Jeganathan [11] (2024)	MODIS, MP & Chhattisgarh, India	Ensemble ML (RF-based)	High accuracy; NDVI & rainfall reduce fire risk	Region-specific; 20-year retrospective data
Muhammad et al. [12] (2025)	MODIS FIRMS/MCD64A1, Pakistan	LR, RF, SVM, XGBoost	RF/XGBoost outperform baselines; NDVI & LST key drivers	Susceptibility, not real-time prediction
Ejaz & Choudhury [14] (2025)	Literature survey	ML pipeline review	Identifies interpretability & generalisation as open gaps	Survey; no new model proposed

4.5 Research Gap Analysis

The literature establishes that deforestation detection via deep-learning segmentation and fire-risk prediction via ensemble machine learning are individually mature research directions, but a structured reading reveals consistent gaps: (i) the overwhelming majority of studies address either change detection or fire-risk prediction, but not both within a single, jointly operated pipeline — even Fodor and Conde [8], who combine deforestation and burned-area detection, do not incorporate pre-ignition fire-risk modelling from environmental variables; (ii) while NDVI is a common fire-risk feature [10]–[12], none of the reviewed deforestation studies feed change-detection outputs forward into a fire-risk model, despite the plausible link between recent disturbance and altered fuel/microclimate conditions; (iii) GIS decision-support tools are largely

built around a single risk layer, and interpretability of a combined risk picture for non-specialist users remains an explicitly noted open challenge [14].

This project does not claim that deforestation detection or fire-risk prediction are themselves unaddressed — both are active, well-populated research areas. The gap identified is the absence of a single implemented pipeline that jointly performs multi-temporal deforestation detection and environmental fire-risk prediction and fuses both outputs into one interpretable, spatially referenced vulnerability visualisation intended for decision support rather than academic benchmarking alone.

5. Research Gap and Project Contribution

Building on Section 4.5, this project's contribution is scoped as an integrated implementation rather than a new neural-network architecture: the deforestation module uses an established encoder–decoder segmentation approach (U-Net or a close variant), and the fire-risk module uses established ensemble techniques (Random Forest and/or XGBoost), both well validated in the literature above.

The contribution lies instead in the design and integration of these components into a single pipeline: multi-temporal imagery is processed through a change-detection model to generate a spatial deforestation map; environmental and historical fire data are processed through a fire-risk model to generate a spatial fire-risk map; and the two outputs are combined through a transparent spatial-fusion procedure into an integrated forest-vulnerability heatmap with an interpretable prototype dashboard — satisfying the college's requirement that the Micro Project demonstrate design, innovation, implementation, or problem-solving, without overstating the novelty of any individual algorithm.

6. Objectives

- To acquire and pre-process multi-temporal Sentinel-2 / Landsat imagery for a selected pilot forest region, including cloud masking, radiometric normalisation, and co-registration across time periods.
- To design and train a deep-learning segmentation model (U-Net or an equivalent encoder–decoder) that detects forest-cover change between time periods and generates a spatial deforestation map.
- To compute percentage forest-cover change and region-wise deforestation statistics from the segmentation output, where dataset resolution and extent permit.
- To collect and harmonise historical fire-occurrence data (e.g., NASA FIRMS), meteorological variables (temperature, humidity, rainfall, wind speed), and NDVI for the same pilot region.
- To develop and evaluate machine-learning models (baseline: Logistic Regression; primary: Random Forest and/or XGBoost) that estimate fire-risk probability and classify it into Low, Moderate, High, and Extreme categories.
- To generate spatial deforestation and fire-risk heatmaps on a consistent grid and fuse them into an integrated vulnerability visualisation using a transparent, explainable scoring procedure.
- To evaluate the segmentation model using IoU, Dice, precision, recall, and F1-score, and the fire-risk model using accuracy, precision, recall, F1-score, and ROC-AUC on held-out data.
- To implement a prototype interactive dashboard displaying the deforestation map, fire-risk heatmap, integrated vulnerability map, and basic temporal-trend visualisation.

7. Proposed Methodology

7.1 Overall System Architecture

The system follows a modular pipeline in which the deforestation-detection and fire-risk-prediction modules operate on independent data sources and converge at a common spatial grid for fusion and visualisation. Satellite imagery is preprocessed and passed through a change-detection model to produce a deforestation map with percentage forest-cover change. In parallel, weather and historical fire data undergo feature engineering and are passed through a

Random Forest / XGBoost fire-risk model to produce a Low/Moderate/High/Extreme fire-risk map. The deforestation map and fire-risk map are then aligned on a common spatial grid within a Risk Fusion Layer that produces an integrated forest-vulnerability heatmap, which feeds an interactive dashboard for decision support.

7.2 Data Acquisition

Satellite imagery will be sourced from Copernicus Sentinel-2 (10–20 m, 5-day revisit) and, for longer baselines, the USGS Landsat archive (30 m, multi-decade coverage). At least two cloud-free scenes of the pilot region, separated by one to three years, will be acquired for change detection. Historical fire-occurrence records will come from NASA FIRMS (MODIS/VIIRS active-fire detections), and meteorological variables (temperature, humidity, rainfall, wind speed) from public reanalysis or station-based archives for the same region and window.

7.3 Data Preprocessing

Satellite scenes will undergo cloud/shadow masking, radiometric normalisation, and geometric co-registration, then be tiled into patches for the segmentation network. Weather and fire records will be joined to a common grid matching the deforestation-output resolution where feasible; missing values handled through interpolation or exclusion, and categorical variables encoded appropriately. NDVI will be computed directly from the red and near-infrared bands of the same imagery used for change detection, ensuring consistency between modules.

7.4 Deforestation Detection Module

A U-Net-style encoder–decoder network, informed by [1]–[8], will classify each pixel of the later-period image (optionally conditioned on the earlier image) as forest, non-forest, or changed, using skip connections to recover precise boundaries. If a labelled multi-temporal dataset is unavailable for the pilot region, the model will first be trained on a public benchmark and fine-tuned via transfer learning on any curated local subset. The resulting mask is post-processed with morphological filtering and converted to a vector polygon layer, from which percentage forest-cover change is computed.

7.5 Fire-Risk Prediction Module

Historical fire points serve as positive labels, with a proportionate sample of non-fire locations/time windows as negatives, following the case-control approach common in [10]–[13]. Feature vectors combine weather variables, NDVI and an NDVI-trend dryness index (following Michael et al. [10]), and static terrain/land-cover attributes. Logistic Regression establishes a baseline; Random Forest and XGBoost — which consistently outperformed simpler baselines in the reviewed studies [10]–[13] — serve as primary models. The continuous fire-probability output is discretised into Low/Moderate/High/Extreme bands calibrated against the predicted-probability distribution and, where possible, cross-validated against known fire locations.

7.6 Heatmap Generation and 7.7 Integrated Vulnerability Assessment

Both module outputs are rasterised onto a common spatial grid using a consistent four-band colour scale (green → yellow → orange → red), with separate layers per time period to support before/after comparison. The Risk Fusion Layer combines the normalised deforestation-change score and fire-risk score per grid cell into a composite vulnerability score via a transparent weighted-sum formulation, with weights refined through sensitivity analysis rather than a black-box combination. Cells with both high recent forest loss and high fire risk are flagged as priority-vulnerability zones, and the composite score is shown alongside — not instead of — the two individual layers so a forest manager can see which factor is driving the result.

7.8 Dashboard

A prototype interactive dashboard will present the deforestation, fire-risk, and integrated vulnerability heatmaps as selectable layers over a base map, together with before/after image panels, a region-wise summary table of forest-loss percentage and dominant risk category, and basic time-series charts. The dashboard is explicitly scoped as a decision-support and visualisation prototype, not an autonomous fire-prevention system.

8. Datasets and Tools

The project will rely exclusively on publicly available, open-access datasets to ensure reproducibility. No dataset statistics are assumed in advance; these will be determined empirically once the pilot region and time window are finalised.

Dataset / Tool	Contents	Relevance to Project
Sentinel-2 (Copernicus)	10–20 m multispectral optical imagery, 5-day revisit, red/NIR bands for NDVI	Primary source for multi-temporal deforestation detection and NDVI computation
Landsat 8/9 (USGS)	30 m multispectral imagery, multi-decade archive	Longer-baseline change detection where Sentinel-2 history is insufficient
NASA FIRMS	Geolocated active-fire detections from MODIS/VIIRS, near-daily	Historical fire-occurrence labels for training/validating the fire-risk model
Public weather / reanalysis archives	Temperature, humidity, rainfall, wind speed	Environmental predictor variables for the fire-risk model
UCI “Forest Fires” dataset (supplementary)	Meteorological and burned-area records for a benchmark region	Optional supplementary dataset for baseline prototyping and validation
QGIS / GDAL	Open-source GIS processing libraries	Raster/vector processing, reprojection, and heatmap rendering
Python (TensorFlow/PyTorch, scikit-learn, XGBoost)	Deep-learning and machine-learning frameworks	Implementation of the segmentation and fire-risk models

9. Model Selection and Evaluation

9.1 Deforestation Detection Models

A baseline pixel-wise classifier (a shallow CNN or a classifier on spectral indices) will establish a lower-bound reference. The primary model is a U-Net-style encoder–decoder, chosen for pixel-accurate segmentation with limited training data; an attention-augmented variant informed by John and Zhang [4] will be evaluated as an enhancement if time and data permit.

9.2 Fire-Risk Prediction Models

Because fire-risk data is fundamentally tabular, classical ensemble ML models are more appropriate than deep learning for this module, consistent with [10]–[13]. Logistic Regression provides an interpretable baseline; Random Forest and XGBoost serve as primary models for their superior reported performance and feature-importance rankings. A recurrent model (e.g., LSTM) will be considered only if sufficiently long temporal weather sequences are available. A comparative evaluation across models will be reported, but this is a supporting step, not the project's central contribution.

9.3 Evaluation Metrics

For deforestation segmentation, IoU and Dice coefficient will be the primary metrics, as they measure spatial overlap and are more informative than raw pixel accuracy under severe class

imbalance; precision, recall, and F1-score will also be reported per class. For fire-risk classification, ROC-AUC, recall, and the confusion matrix will be given particular weight, since a false negative (a high-risk area misclassified as low-risk) is materially more costly than a false positive in an early-warning context.

Component	Primary Metrics	Why These Matter
Deforestation segmentation	IoU, Dice coefficient, Precision, Recall, F1-score	Robust to severe class imbalance between changed and unchanged pixels; pixel accuracy alone would be misleading
Fire-risk classification	Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix	ROC-AUC and recall are prioritised over raw accuracy since missed high-risk cases are more costly than false alarms

10. Expected Outcomes

- A trained deep-learning model generating a forest-cover change / deforestation map for the pilot region from multi-temporal satellite imagery, with quantified deforestation statistics.
- A trained machine-learning model outputting a fire-risk probability and Low/Moderate/High/Extreme category for each grid cell.
- Spatial fire-risk and deforestation heatmaps rendered on a consistent grid and colour scale, and an integrated forest-vulnerability heatmap combining both through a transparent fusion procedure.
- A prototype interactive dashboard displaying all layers with before/after imagery and basic time-series visualisation.
- A quantitative evaluation of both models against the metrics in Section 9.3, reported on held-out data — no specific accuracy, IoU, or AUC figures are presumed in advance; all figures will be determined experimentally and reported honestly, including any limitations observed.

11. Novelty and Innovation

As established in Section 5, this project's novelty lies in system design and integration rather than a new neural-network architecture. It contributes an integrated framework that jointly operates deep-learning deforestation detection and machine-learning fire-risk prediction within a single pipeline; a multi-temporal change-detection approach applied consistently alongside the

fire-risk model on the same pilot region and grid; a transparent, explainable risk-fusion procedure that preserves visibility into each contributing layer, addressing the interpretability gap noted in [14]; and a decision-support dashboard presenting all outputs in one interface for a non-specialist forest manager or researcher — satisfying the college's evaluation criteria while remaining realistic in scope for a B.Tech CSE Micro Project.

12. SDG Alignment

SDG 13 — Climate Action

Forest fires and deforestation are major contributors of land-use greenhouse-gas emissions. By generating pre-ignition fire-risk assessment rather than only reacting to active fires, the system supports preventive action — pre-positioning resources, scheduling controlled burns, issuing advisories — that can reduce the frequency and scale of large fire events and their emissions.

SDG 15 — Life on Land

Continuous, low-cost monitoring of forest-cover change enables earlier detection of illegal logging and degradation than manual surveys, supporting timely conservation intervention, with the integrated vulnerability view helping prioritise effort toward areas facing both active forest loss and elevated fire susceptibility.

Secondary Relevance

SDG 11 (Sustainable Cities and Communities) is relevant where fire-risk zones lie near human settlements, since early-warning information can inform disaster preparedness at the wildland–urban interface. SDG 6 (Clean Water and Sanitation) has a plausible but indirect connection, since forest loss and severe fires degrade watershed function; this project does not model hydrological outcomes directly and treats SDG 6 as contextual rather than core.

13. Practical Applicability

The prototype is designed to be useful to several categories of users, while remaining explicitly scoped as a decision-support tool rather than an autonomous or emergency-response system:

- Forest departments and range officers, to prioritise patrolling and enforcement in high-loss or high-risk zones.
- Disaster-management agencies, to use the fire-risk heatmap as one input for pre-season resource planning and public advisories.
- Environmental researchers and conservation organisations, to use the outputs as a starting point for field-level investigation of high-vulnerability regions.
- Academic researchers, for whom the open, reproducible pipeline could serve as a foundation for further methodological extensions.

This system is a prototype and decision-support aid, not a replacement for official emergency-response systems, certified fire-detection services, or statutory forest-survey mechanisms; any operational deployment would require validation against ground-truth data and sign-off from relevant domain authorities.

14. Limitations

- Cloud cover and atmospheric interference can obscure optical imagery, particularly in tropical and monsoon-affected regions, reducing usable scenes for change detection.
- Labelled deforestation data for the specific pilot region may be limited, potentially requiring transfer learning from other geographies, which can affect segmentation accuracy.
- Models trained on one geographic region may not generalise well to forests with different vegetation, climate, or fire regimes without retraining.
- The fire-risk model will produce both false positives and false negatives; risk categories must be interpreted as probabilistic guidance, not certainties.
- Satellite resolution (10–30 m) and coarser weather-data resolution may not capture fine-grained, localised variation relevant at very small scales.
- Historical weather and fire-occurrence data may be incomplete for some regions and periods, constraining training and evaluation; and establishing causality between

deforestation and subsequent fire is difficult with observational data alone — the project treats the two as correlated indicators of vulnerability rather than asserting direct causation.

15. Future Scope

- Incorporating near-real-time satellite feeds and automated retraining pipelines for continuous rather than periodic monitoring.
- Integrating drone/UAV imagery for higher-resolution validation of AI-flagged high-vulnerability zones.
- Exploring spatiotemporal architectures (e.g., ConvLSTM or transformers) that jointly model spatial and temporal patterns across both signals.
- Adding explainable-AI techniques (e.g., SHAP, Grad-CAM) for per-prediction feature attribution in both modules.
- Integrating low-cost IoT ground sensors to supplement satellite-derived features with localised measurements.
- Extending the dashboard with mobile push-alerts for field staff, and formal integration with official GIS/forest-survey platforms subject to validation and institutional review.
- Publishing the methodology and pilot-region results as a peer-reviewed paper, as encouraged by the college's Micro Project guidelines.

16. References

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